Computational Pathology: Data, Preprocessing, Model Training

Seminar ML in Medicine, Skill Session, 21/11/2025

Julius Hense

Computational Pathology: Data, Preprocessing, Model Training

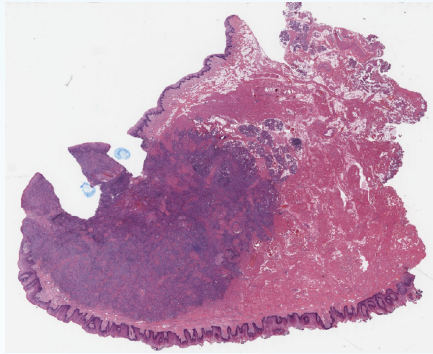
Overview

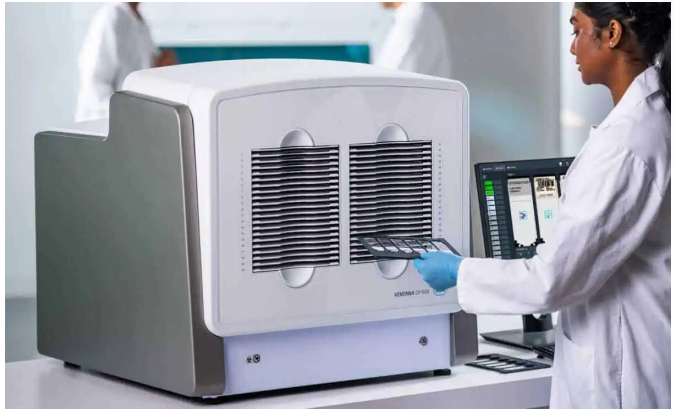
1. Slide Digitization
2. WSI Preprocessing
3. Training Paradigms
4. Types of Papers

Goal: Convey the “basics” of the field to make understanding the papers easier.

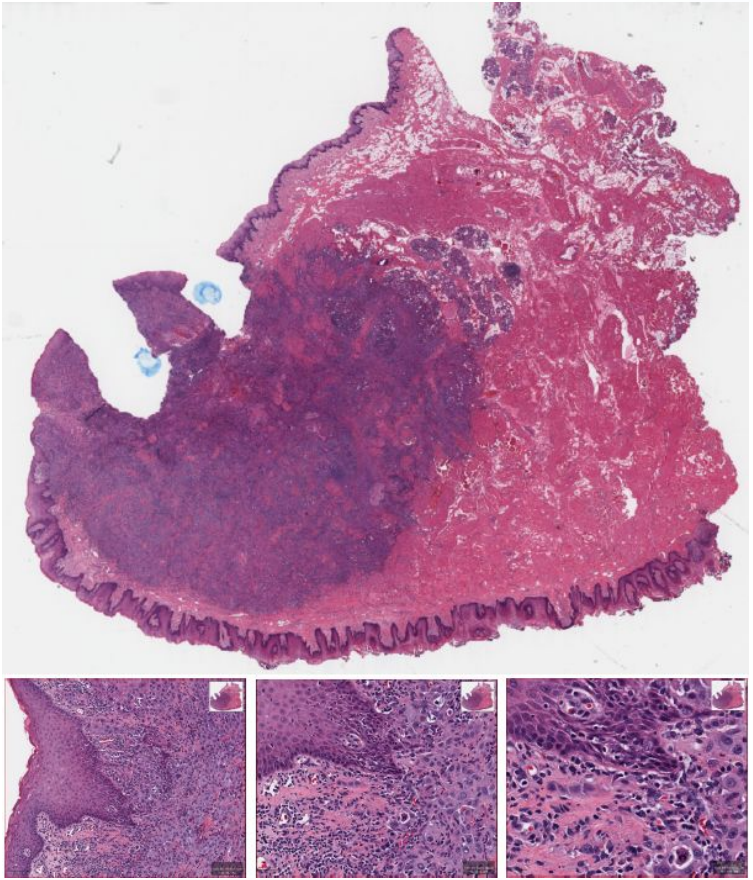
Digitization of Tissue Slides

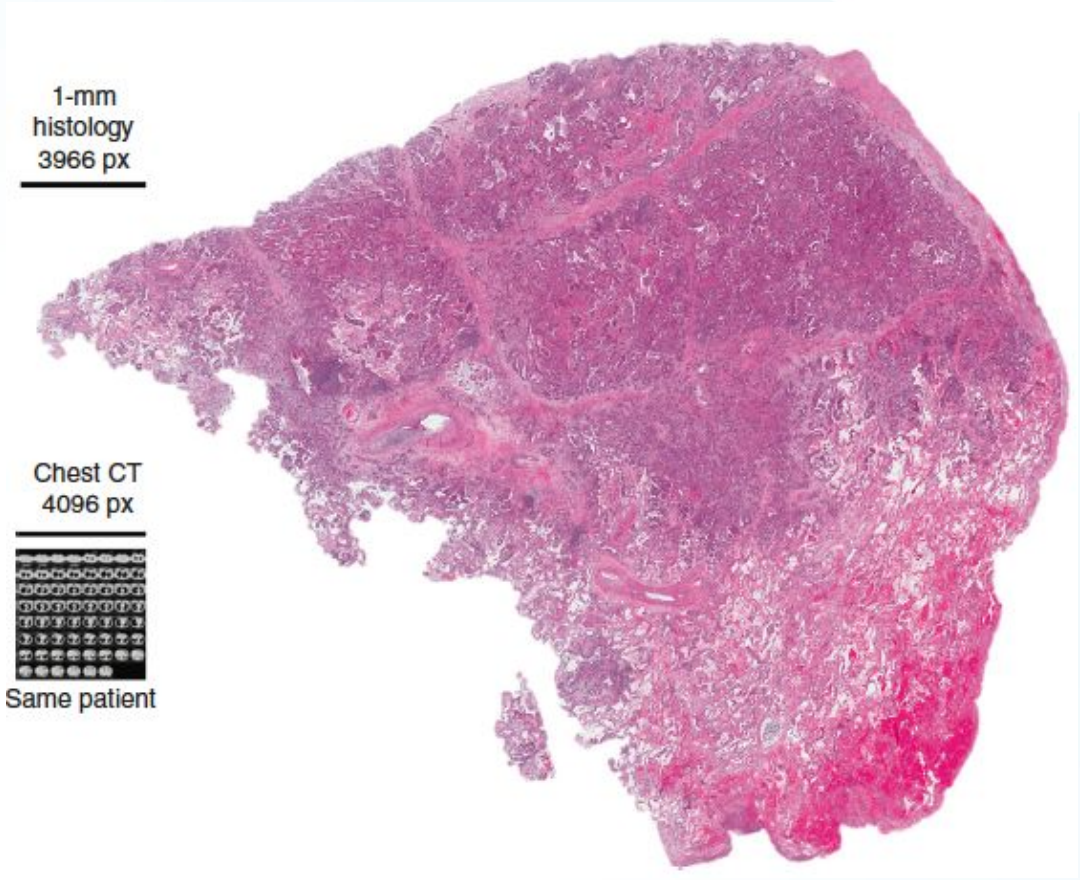
BIFOLD





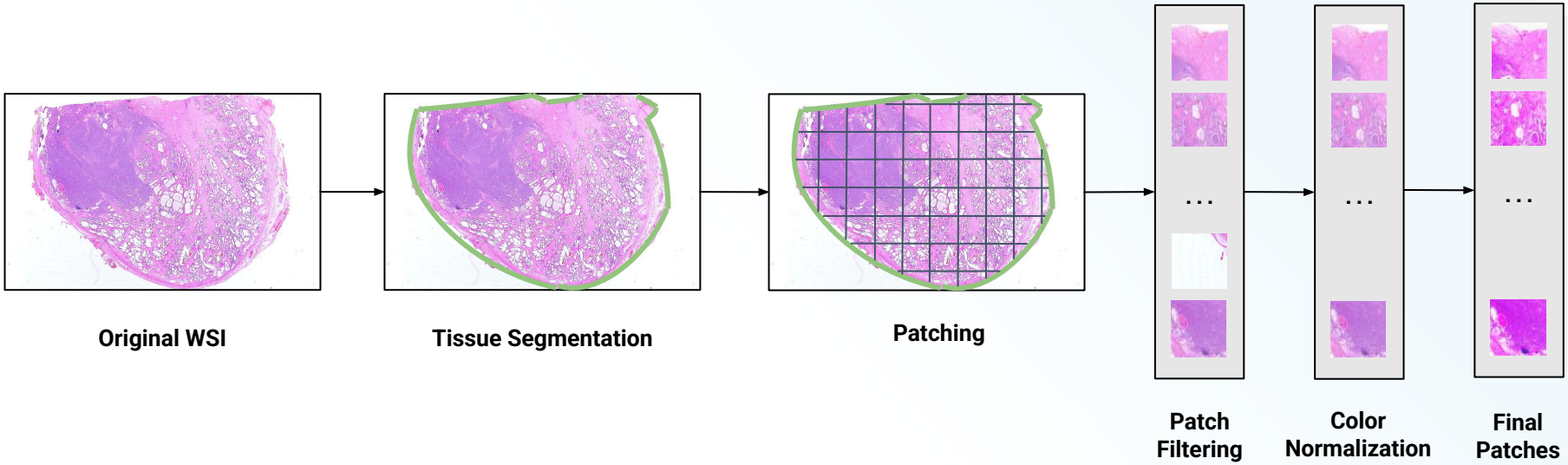
Digital Whole Slide Images (WSIs)



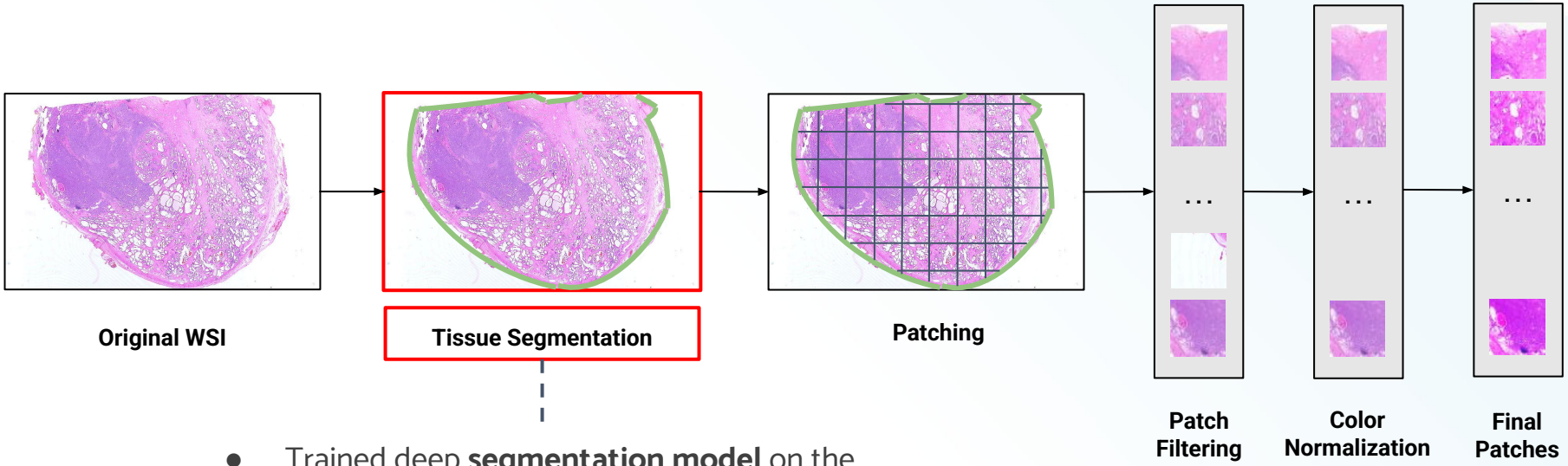


From: Echle, Amelie, et al. "Deep learning in cancer pathology: a new generation of clinical biomarkers." *British journal of cancer* 124.4 (2021): 686-696.

WSI Preprocessing for ML Pipelines

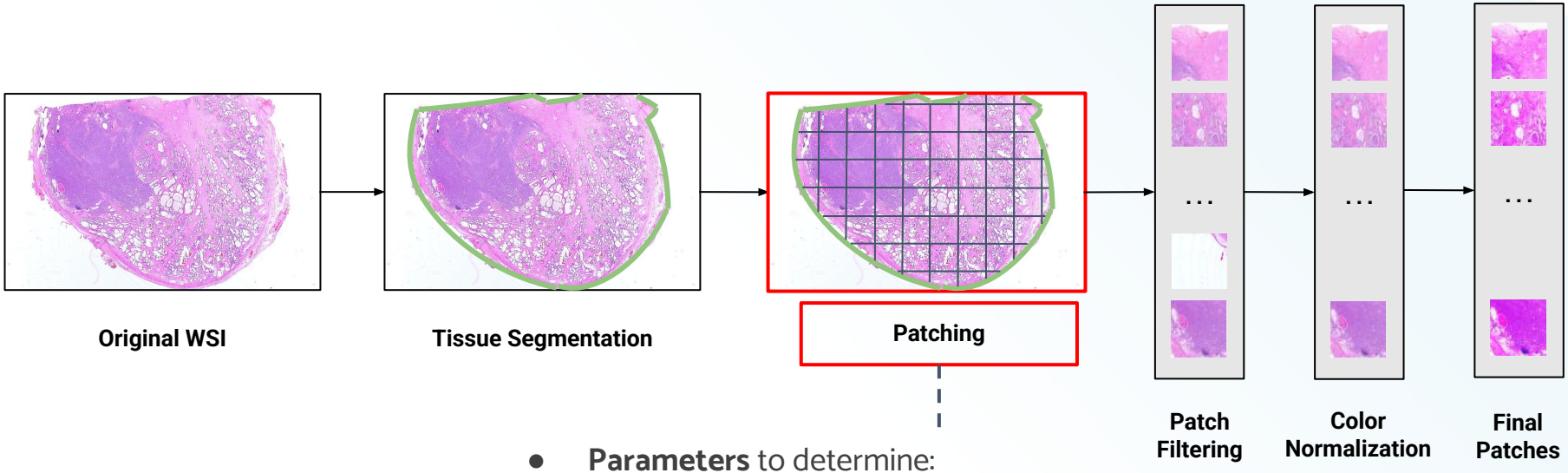


WSI Preprocessing for ML Pipelines



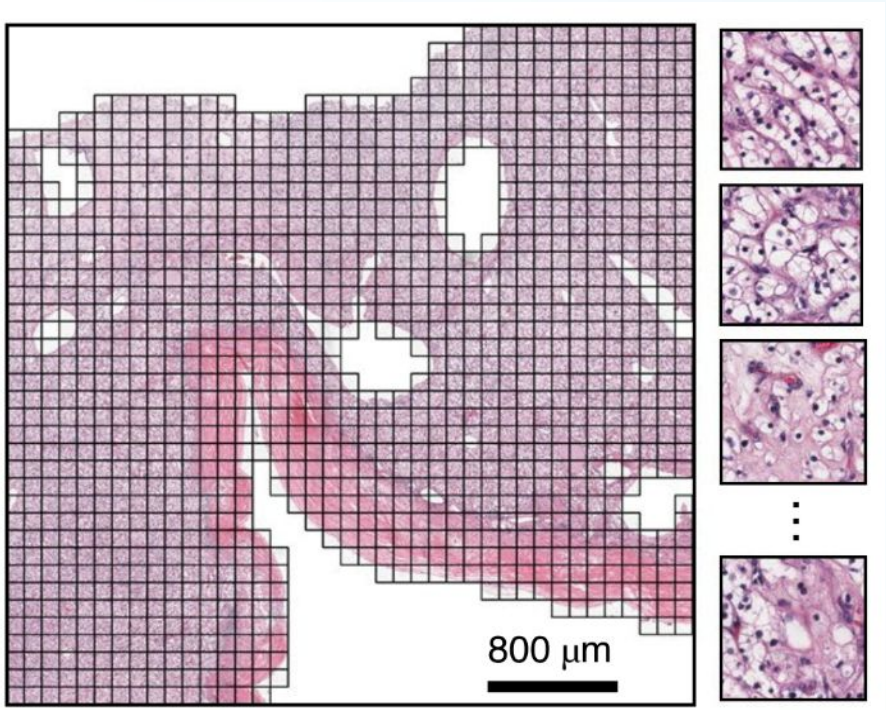
- Trained deep **segmentation model** on the slide thumbnail or on patches.
- Traditional **computer vision methods** on the slide thumbnail (e.g. Otsu's method).

WSI Preprocessing for ML Pipelines



- **Parameters** to determine:
  - Magnification (e.g. 40x, 20x, 10x, 5x)
  - Patch size (in pixel)
  - Patch overlap
- Typical example: 20x, 256x256 pixels, no overlap; often results in **>10K patches per slide**

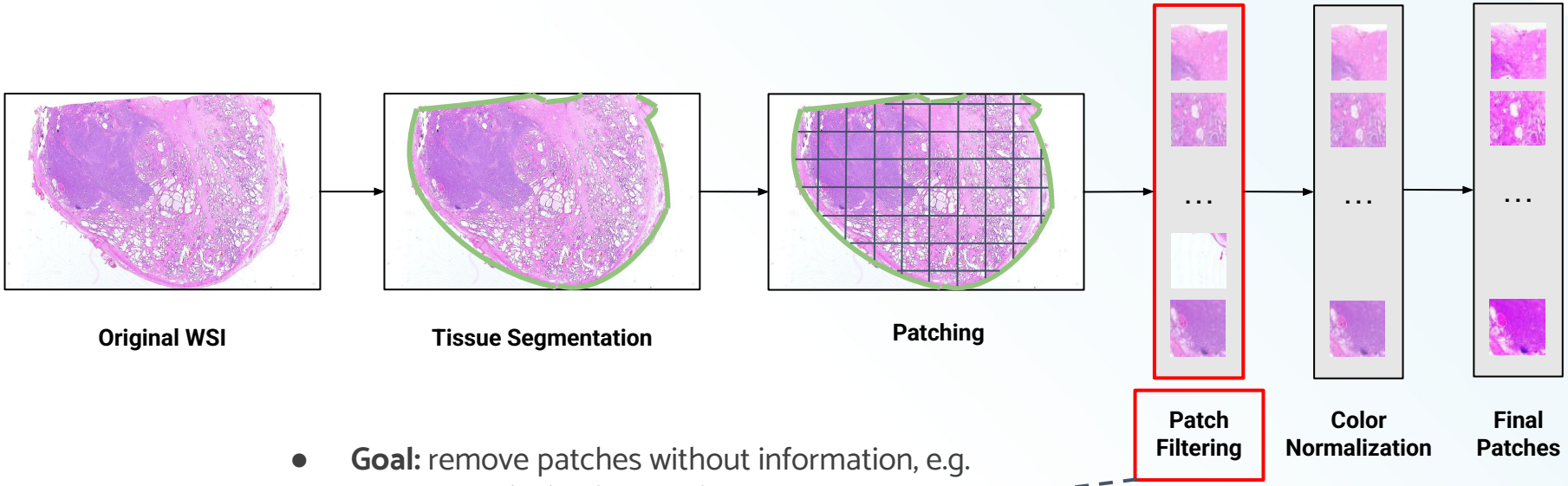
WSI Preprocessing for ML Pipelines



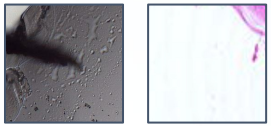
From: Lu, Ming, et al. "Data-efficient and weakly supervised computational pathology on whole-slide images." *Nature biomedical engineering* 5.6 (2021): 555-570.

WSI Preprocessing for ML Pipelines

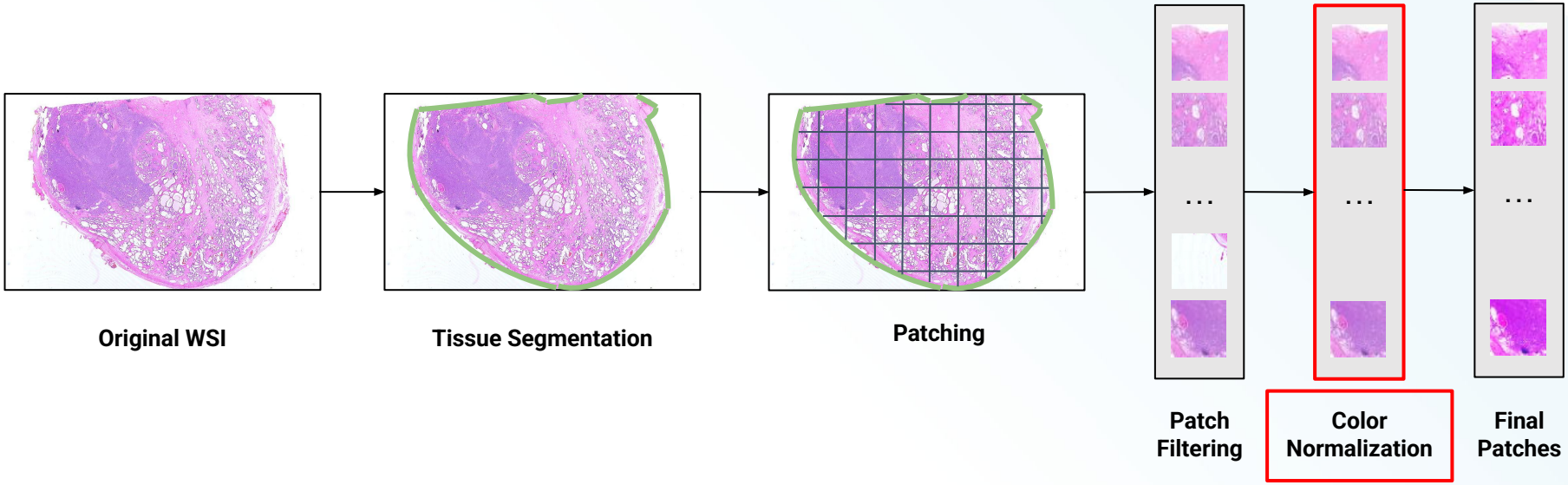
BIFOLD

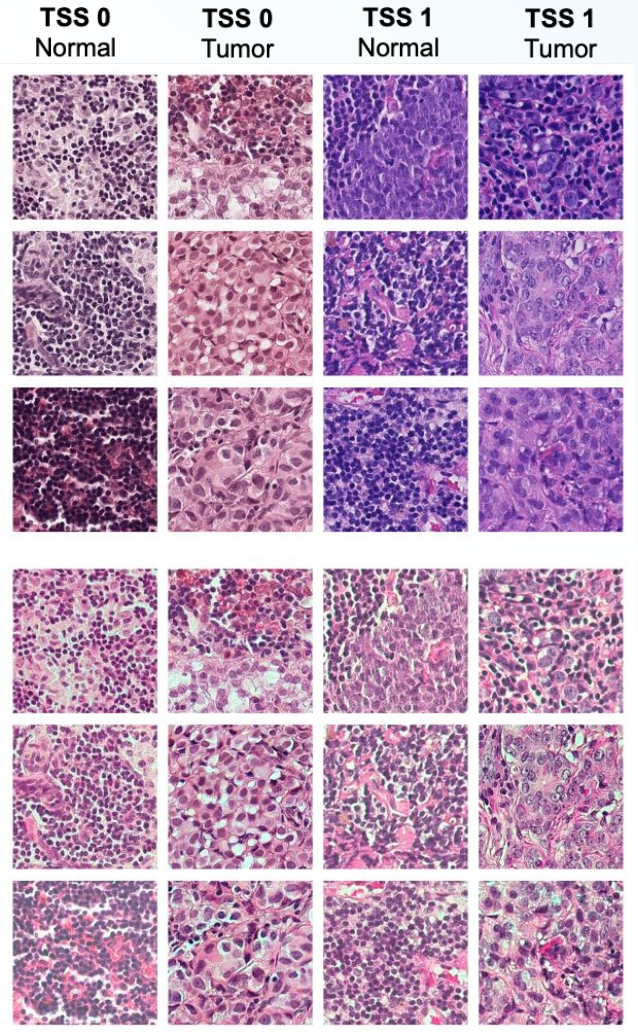


- **Goal:** remove patches without information, e.g.
  - (Mostly) background
  - Artifacts (dirt, blurring, pen mark, ...)
- Trained artifact detection models.
- Traditional computer vision methods (e.g. standard deviation filter).



WSI Preprocessing for ML Pipelines





Raw patches

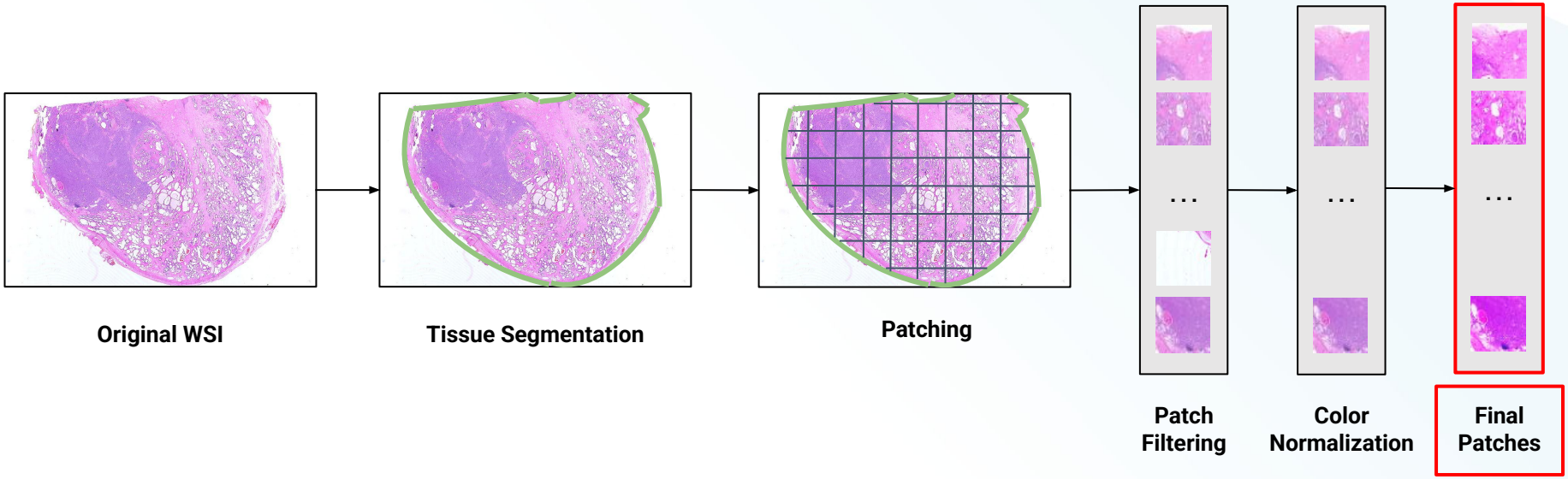
Reinhard color normalization

Frequently used **methods**:

- Reinhard color normalization [1]
- Macenko color normalization [2]
- Advanced color normalization [3]
- Color augmentations during training

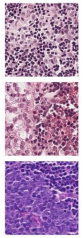
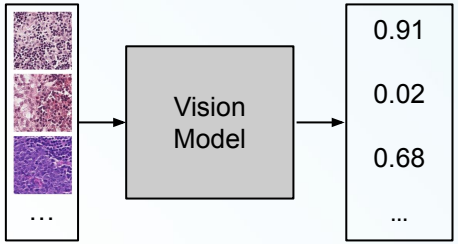
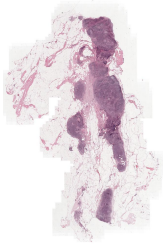
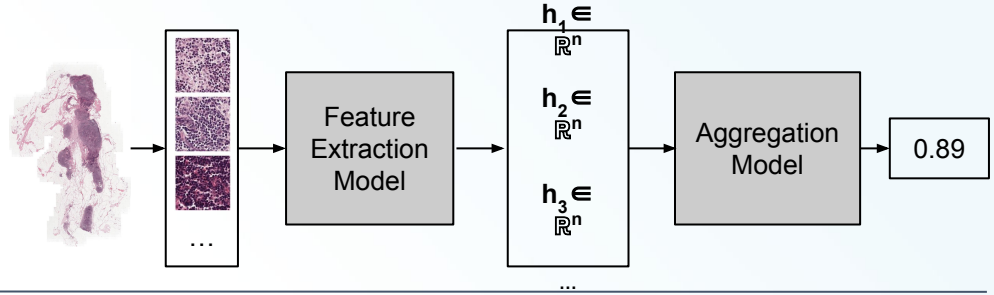
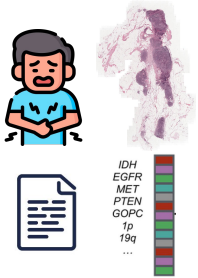
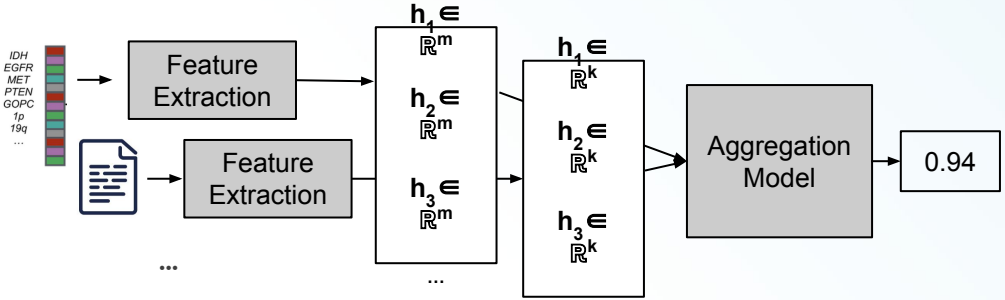
[1] Reinhard, Erik, et al. "Color transfer between images." IEEE Computer graphics and applications 21.5 (2001): 34-41.
[2] Macenko, Marc, et al. "A method for normalizing histology slides for quantitative analysis." 2009 IEEE international symposium on biomedical imaging: from nano to macro. IEEE, 2009.
[3] Hoque, Md Ziaul, et al. "Stain normalization methods for histopathology image analysis: A comprehensive review and experimental comparison." Information Fusion 102 (2024): 101997.

WSI Preprocessing for ML Pipelines

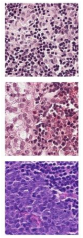
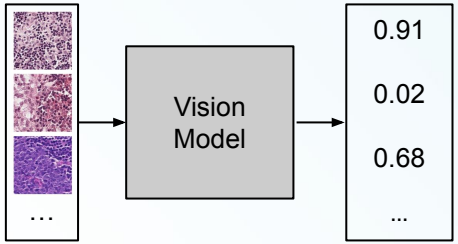
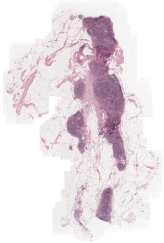
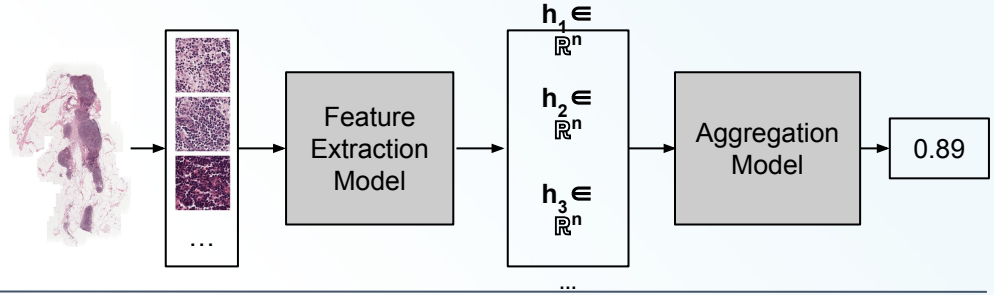
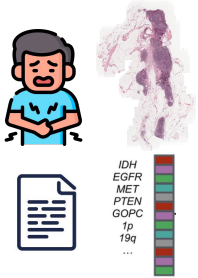
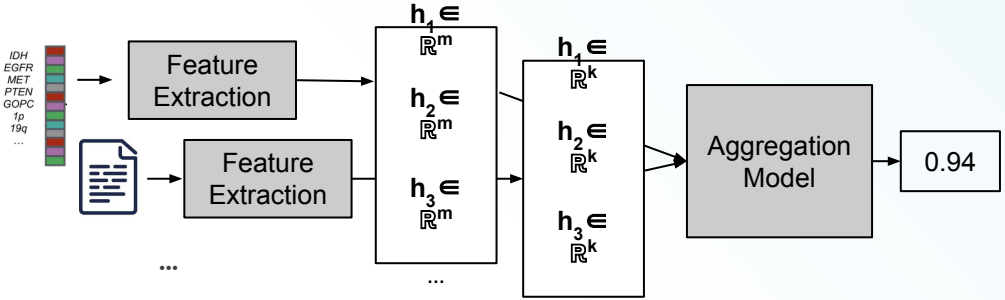


Training Paradigms in Computational Pathology

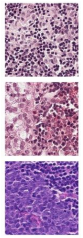
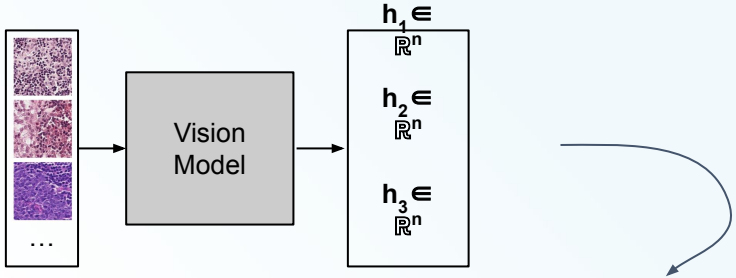
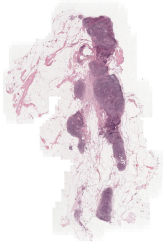
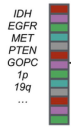
BIFOLD

Paradigm	Data	Labels	Pipeline
Patch-level supervision		This is a tumor patch. This is a normal patch. This is a tumor patch.	
Slide-level supervision		This slide contains tumor cells somewhere.	
Patient-level supervision		This patient responded to a treatment.	

Training Paradigms in Computational Pathology

Paradigm	Data	Labels	Pipeline
Supervised training		This is a tumor patch. This is a normal patch. This is a tumor patch.	
Weakly supervised training		This slide contains tumor cells somewhere.	
Weakly supervised training		This patient responded to a treatment.	

Training Paradigms in Computational Pathology

Paradigm	Data	Labels	Pipeline
Self-supervised training		- - -	
Self-supervised training		-	
Self-supervised training		-	

From: Ciga, Ozan, et al. "Self supervised contrastive learning for digital pathology." *Machine Learning with Applications* 7 (2022): 100198.

Types of Papers in Computational Pathology

BIFOLD

The "Methodical" Paper

- **Focus**: New / improved ML methods
- **Main audience**: Machine learning experts
- **Structure**: Method first, experiments afterwards
- **Novelty**: The method
- **Published in**: ML conferences or journals (e.g. *NeurIPS*, *CVPR*, *ICML*)
- **Advice for presentation**: Focus on main contribution

The "Medical" Paper

- **Focus**: Biomedical insights / clinical utility
- **Main audience**: Medical professionals
- **Structure**: Results first, methods in appendix
- **Novelty**: The data & insights
- **Published in**: Medical journals (e.g., *Nature Medicine*, *Cancer Cell*, *JAMA*)
- **Advice for presentation**: Read methods section first & focus on main contribution

Warning: Many papers are a mix of these two archetypes with varying degrees of methodical and medical novelty. They are often published in interdisciplinary conferences or journals, e.g. *Medical Image Analysis*, *Nature Biomedical Engineering*, *NEJM AI*, *MICCAI*.

Types of Papers in Computational Pathology

BIFOLD

- **Methodological studies**
  - Demonstrate clear innovations and contributions over the state of the art methodologies.
  - Evaluation and performance assessment is potentially limited to proof of concepts or small-size validation studies.

- **Application studies including clinical translation**
  - Demonstrate clear clinical value of existing techniques, or adoption of state-of-the-art methods to a new problem or context, with appropriate and rigorous evaluation design
  - Do not necessarily need to involve fundamental methodological innovations

From: MICCAI 2025 Reviewer Orientation Slides (https://conferences.miccai.org/2025/en/INSTRUCTIONS-TO-REVIEWERS.html).

Questions?